Cold Storage Temperature Monitoring System

A smart IoT-based solution for real-time temperature monitoring and automated alerts in agricultural cold storage facilities across Uttar Pradesh

SR UNIVERSITY

Current Challenges in Manual Monitoring

Manual Readings

Workers check temperatures every few hours using thermometers, leading to inconsistent data collection and potential human errors in recording

Lost Data

Temperature readings written in notebooks can be misplaced, damaged, or inaccurately transcribed, creating gaps in historical records

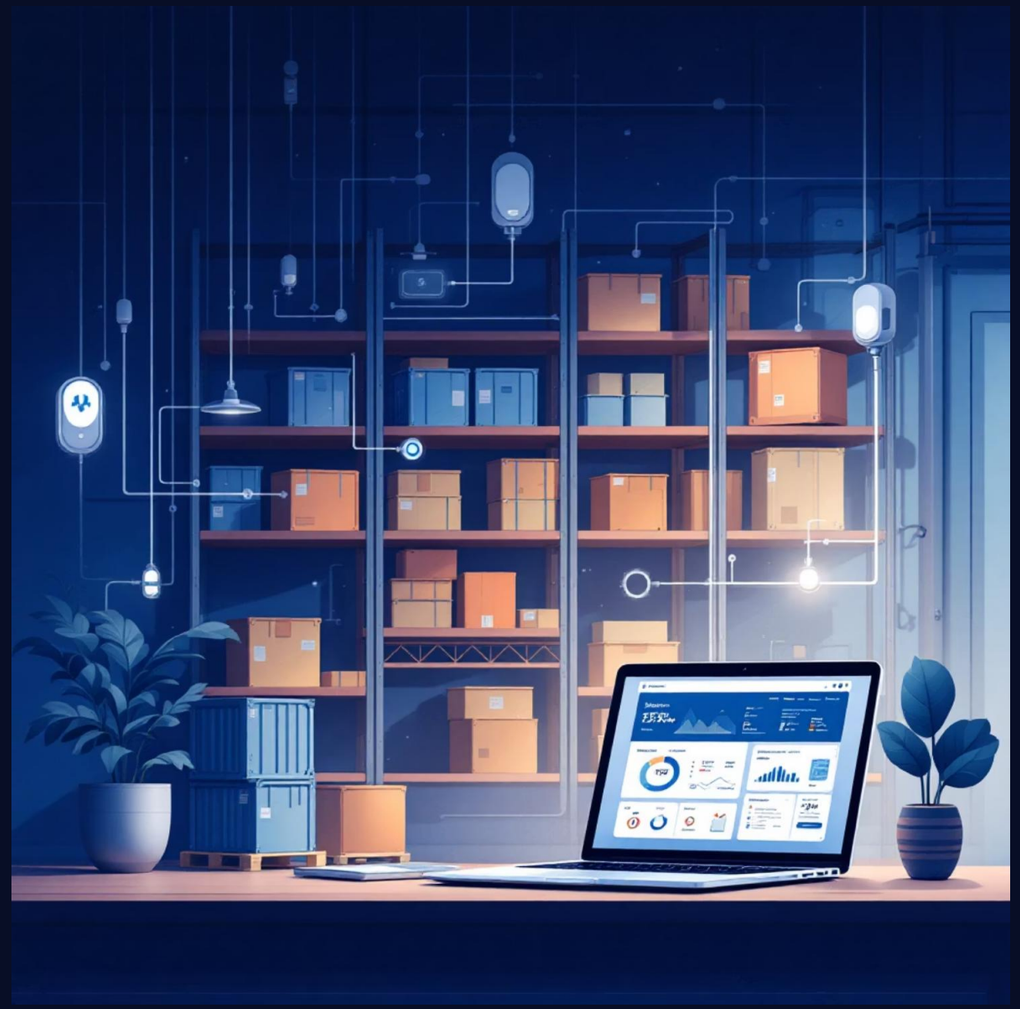
No Alerts

When temperature rises suddenly due to power failures or equipment issues, there is no automatic notification system to warn workers

Financial Loss

Delayed response to temperature changes causes spoilage of potatoes and vegetables, resulting in significant financial losses for farmers and storage owners

Proposed Solution Overview



Smart Monitoring System

A cost-effective IoT-based system that continuously monitors temperature, records data automatically, and sends instant alerts when abnormal conditions are detected.

The solution enables remote monitoring through a web dashboard, eliminating the need for manual checks and ensuring real-time visibility into storage conditions.

System Architecture

01

Temperature Sensors

Digital sensors placed throughout storage units continuously measure temperature

02

Data Gateway

Microcontroller collects sensor data and transmits to cloud server

03

Cloud Platform

Central server stores data, processes alerts, and manages user access

04

User Interface

Web dashboard displays real-time data, historical trends, and alert notifications

Core Functional Requirements

Continuous Monitoring

Real-time temperature tracking with configurable measurement intervals

Data Recording

Automated logging of temperature readings with timestamps and storage unit identification

Alert System

Instant notifications via SMS and email when temperature exceeds threshold values

Daily Reports

Automated generation of comprehensive reports showing temperature trends and anomalies

Power Tracking

Detection and logging of power failures with duration and impact analysis

Multi-Unit Support

Simultaneous monitoring of multiple cold storage facilities from single dashboard

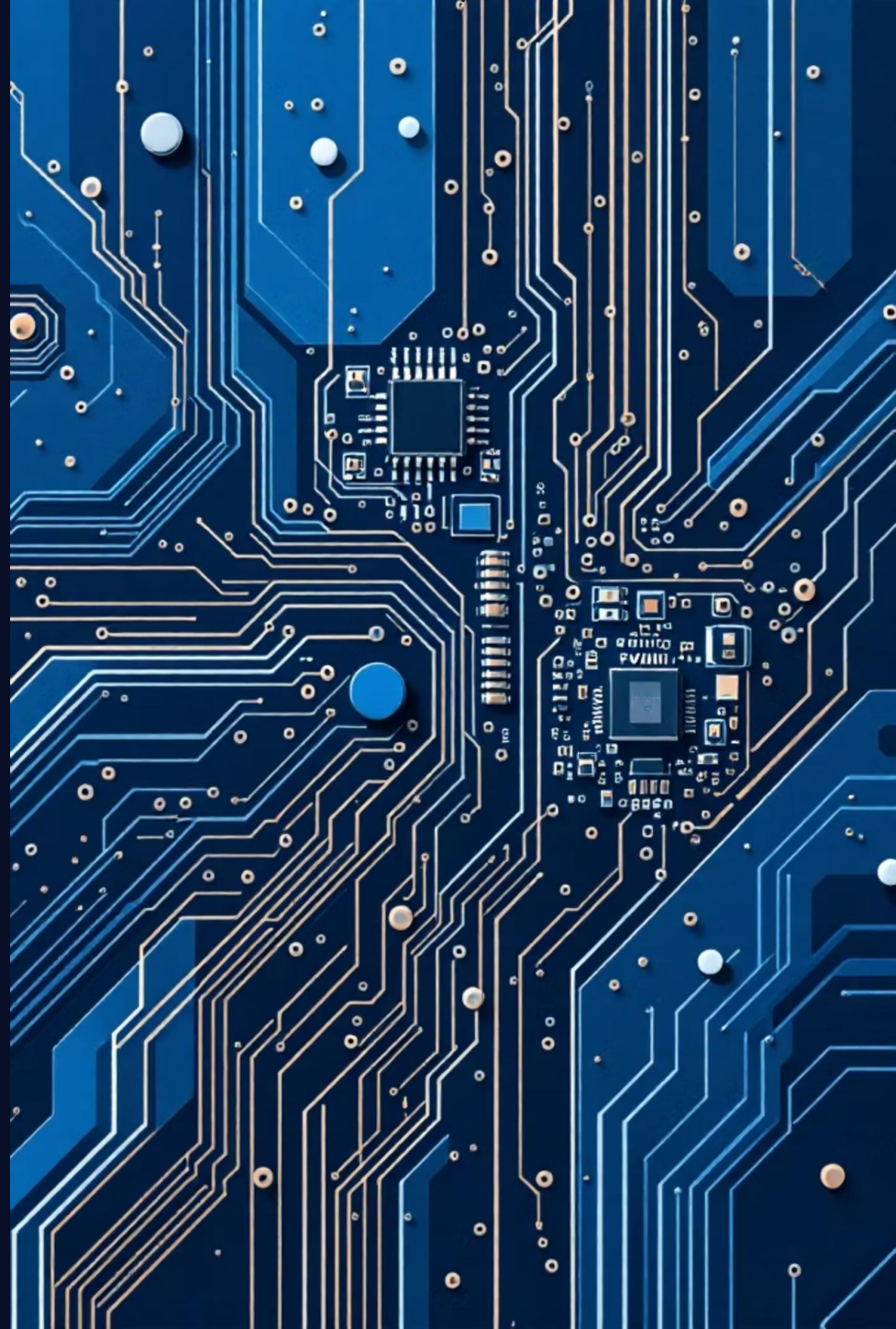
Technical Stack & Implementation

Hardware Components

- DHT22 temperature sensors for accurate readings
- Arduino or ESP32 microcontroller for data collection
- GSM module for SMS notifications
- Power backup system for uninterrupted monitoring

Software Stack

- Python/Node.js for backend server
- MySQL/PostgreSQL for data storage
- React.js for responsive web interface
- Twilio API for SMS alerts



Dashboard & User Interface

Real-Time Display

Live temperature readings with color-coded indicators showing normal, warning, and critical levels

Maintenance Log

Storage managers can record equipment maintenance activities and track service schedules

Historical Analytics

Graphical visualization of temperature trends over time with comparison tools and anomaly detection

Alert Notification System

Detection

System identifies temperature deviation from set parameters

Processing

Server validates anomaly and determines alert priority

Distribution

Simultaneous SMS and email notifications sent to multiple contacts

Benefits & Impact

75%

Reduced Spoilage

Timely alerts prevent produce loss from temperature fluctuations

100%

Data Accuracy

Eliminates manual errors with automated digital recording

24/7

Continuous Monitoring

Round-the-clock surveillance without human intervention

Cost Savings

Reduces financial losses from spoiled produce and optimizes energy consumption through better temperature management

Scalability

System can be deployed across multiple storage facilities with minimal additional cost per unit

Future Enhancements

1

Phase 1

Basic temperature monitoring with SMS alerts and daily reports

2

Phase 2

Integration with humidity sensors and door open/close detection

3

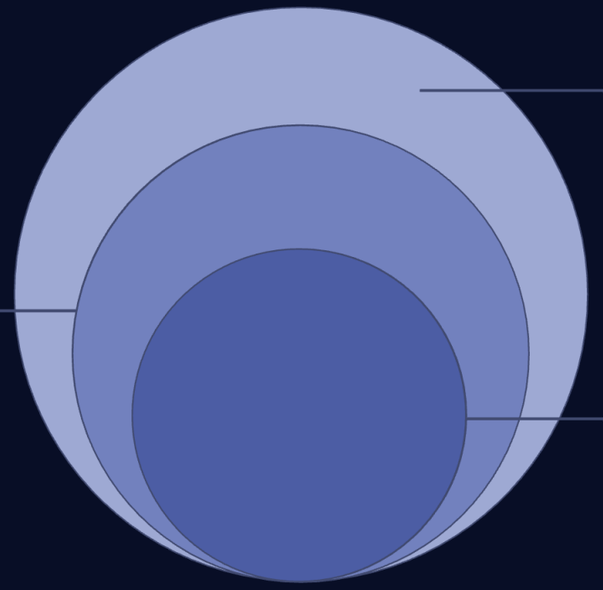
Phase 3

Predictive analytics using machine learning to prevent issues

4

Phase 4

Mobile app with voice commands and integration with agricultural market platforms



Benefits & Future

Reduced spoilage, accuracy, scalability, and phased enhancements

Notifications

Simultaneous SMS and email to multiple contacts

Core

Server validates anomalies and sets priority